Stochastic Models and Simulation

Week 2: how limited service capacity creates waiting

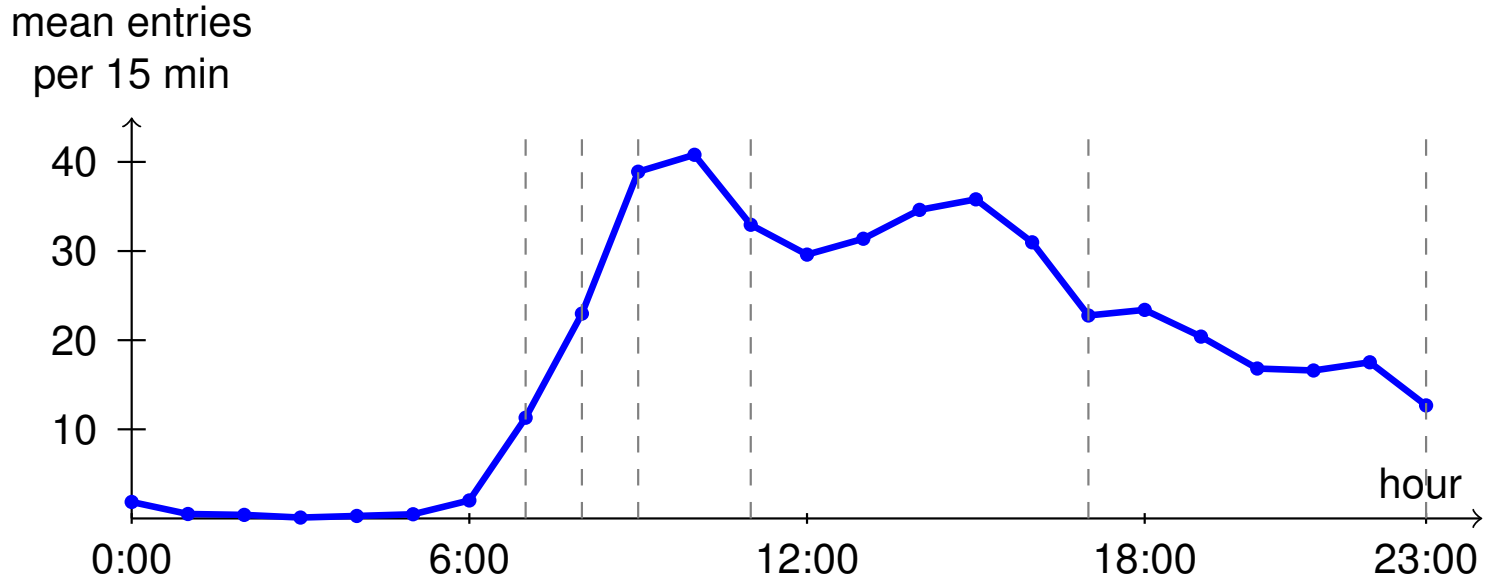
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Recap: arrival times

Training daily mean: Sun–Thu 1,781; Friday 539; Saturday 247.



The curve uses ten **regular** working days (Sun–Thu) from 1–14 November. Each point averages $4 \times 10 = 40$ quarter-hour counts. Twelve later working days are held out.

- Use separate profiles for regular working days, Friday and Saturday.
- On the training plot, choose simple periods by eye. Treat the boundaries as provisional and check them on held-out days.

Candidate 1: a pooled Poisson profile

The dashed boundaries on the training plot define seven periods:

Period b	Clock time	Fitted mean $\widehat{m}_b$ per 15 min
Night	00:00–07:00	0.8
Opening	07:00–08:00	11.3
Build-up	08:00–09:00	23.0
Morning peak	09:00–11:00	39.9
Daytime	11:00–17:00	32.6
Evening	17:00–23:00	19.6
Late	23:00–24:00	12.7

For a slot s in period b, start with

$$N_{d,s} \sim \text{Poisson}(\widehat{m}_b).$$

This model uses seven fitted means instead of one before.

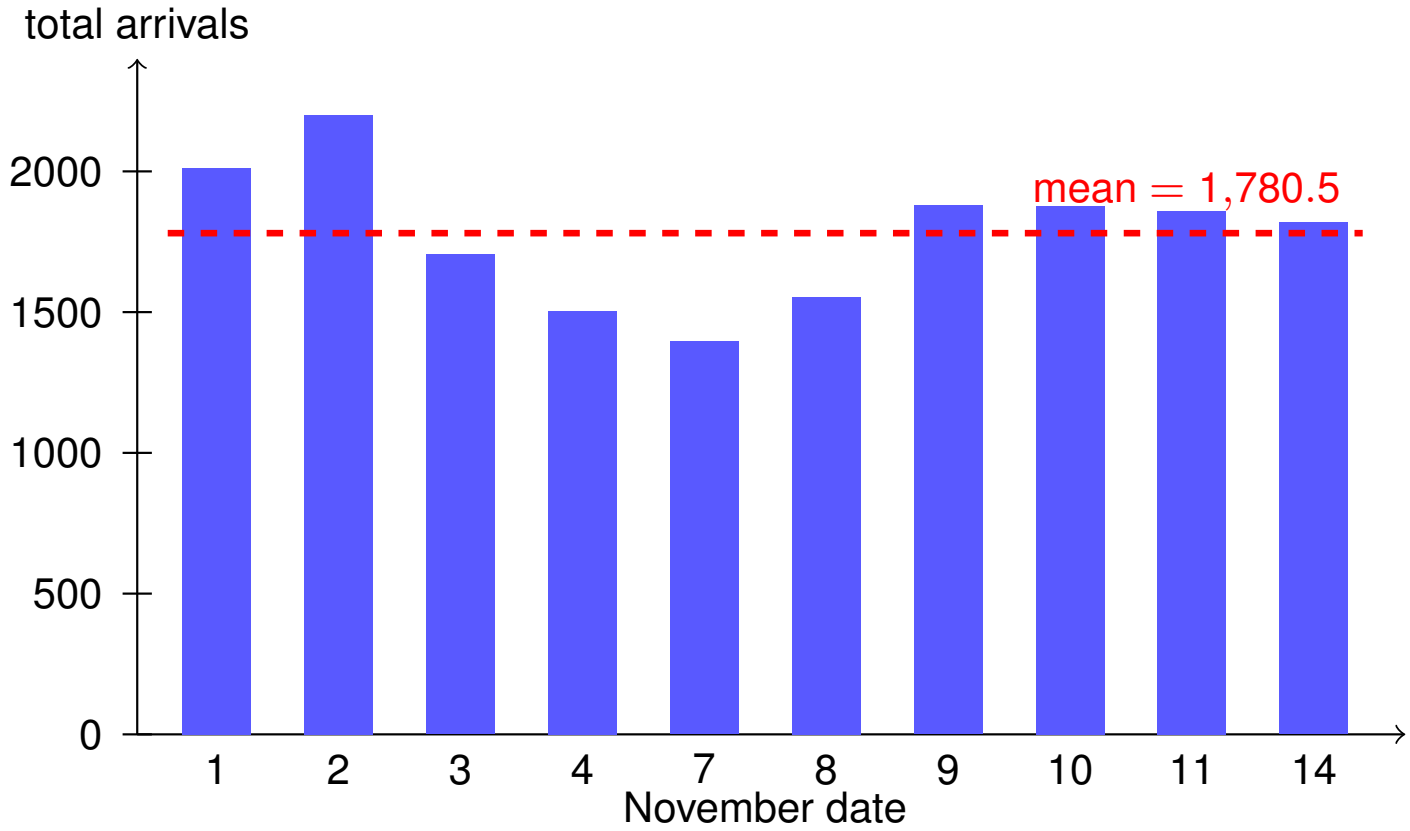
Check the model on new days

The check set contains 12 regular working days from 15–30 November, which were not used to fit the model.

	Mean		Variance	
Period	Fitted	Held out	Poisson	Held out
Night	0.8	0.5	0.8	0.8
Opening	11.3	8.2	11.3	9.2
Build-up	23.0	19.4	23.0	42.7
Morning peak	39.9	38.6	39.9	67.8
Daytime	32.6	31.0	32.6	55.0
Evening	19.6	18.6	19.6	73.8
Late	12.7	11.6	12.7	21.8

Busy-period means are reasonably close, but the fixed-rate Poisson variances are too small.

Daily demand varies across regular working days



Each bar is the total number of arrivals on one complete regular working day in the training period. The totals range from 1,396 to 2,200: some days are quieter or busier throughout the day.

Candidate 2: one daily level, many interval counts

For each regular working day d:

1. Draw one daily demand multiplier G_d. A value above 1 means a busier day; a value below 1 means a quieter day.
2. Let s denote one 15-minute slot and let $b(s)$ be the period containing that slot. The fitted baseline mean $\widehat{m}_{b(s)}$ comes from Candidate 1.
3. Generate a separate arrival count for every slot:

$$N_{d,s} \mid G_d \sim \text{Poisson}\left(G_d \widehat{m}_{b(s)}\right).$$

Example: At 09:30, $b(s)$ is the morning peak and $\widehat{m}_{b(s)} = 39.9$. If $G_d = 1.10$, the Poisson mean for that slot is $1.10 \times 39.9 = 43.9$.

Fit the amount of day-to-day variation

Let the total arrivals on day d be

$$T_d = \sum_{s=1}^{96} N_{d,s}, \qquad M = \sum_{s=1}^{96} \widehat{m}_{b(s)} = 1{,}780.5.$$

If the slot counts are conditionally independent, their sum satisfies

$$T_d \mid G_d \sim \text{Poisson}(G_d M).$$

Set $\mathbb{E}[G_d] = 1$ and $\text{Var}(G_d) = \alpha$. Then

$$\mathbb{E}[T_d] = M, \qquad \text{Var}(T_d) = \underbrace{M}_{\text{Poisson variation}} + \underbrace{\alpha M^2}_{\text{variation between days}}.$$

The ten training totals have sample mean 1,780.5 and sample variance 59,632.7. Matching the model variance to the sample variance gives

$$\widehat{\alpha} = \frac{59{,}632.7 - 1{,}780.5}{1{,}780.5^2} = 0.01825.$$

Choose a distribution that we can simulate

We need a positive multiplier with mean 1 and variance $\widehat{\alpha}$. For

$$G_d \sim \text{Gamma}(k, \theta),$$

using the shape–scale convention,

$$\mathbb{E}[G_d] = k\theta, \qquad \text{Var}(G_d) = k\theta^2.$$

Choosing

$$\theta = \widehat{\alpha} = 0.01825, \qquad k = \frac{1}{\widehat{\alpha}} = 54.8$$

gives the required mean and variance. The standard deviation of G_d is

$$\sqrt{0.01825} = 0.135,$$

so day-level demand varies by about 13.5% of its mean per standard deviation.

Held-out check: the second model is credible for the peak

Each row again pools 15-minute counts. Sample variances use divisor $n - 1$.

	Mean		Variance	
Period	Model	Held out	Model	Held out
Night	0.8	0.5	0.8	0.8
Opening	11.3	8.2	13.6	9.2
Build-up	23.0	19.4	32.6	42.7
Morning peak	39.9	38.6	68.8	67.8
Daytime	32.6	31.0	51.9	55.0
Evening	19.6	18.6	26.6	73.8
Late	12.7	11.6	15.6	21.8

What did the two arrival models show?

Candidate 1: fixed rates

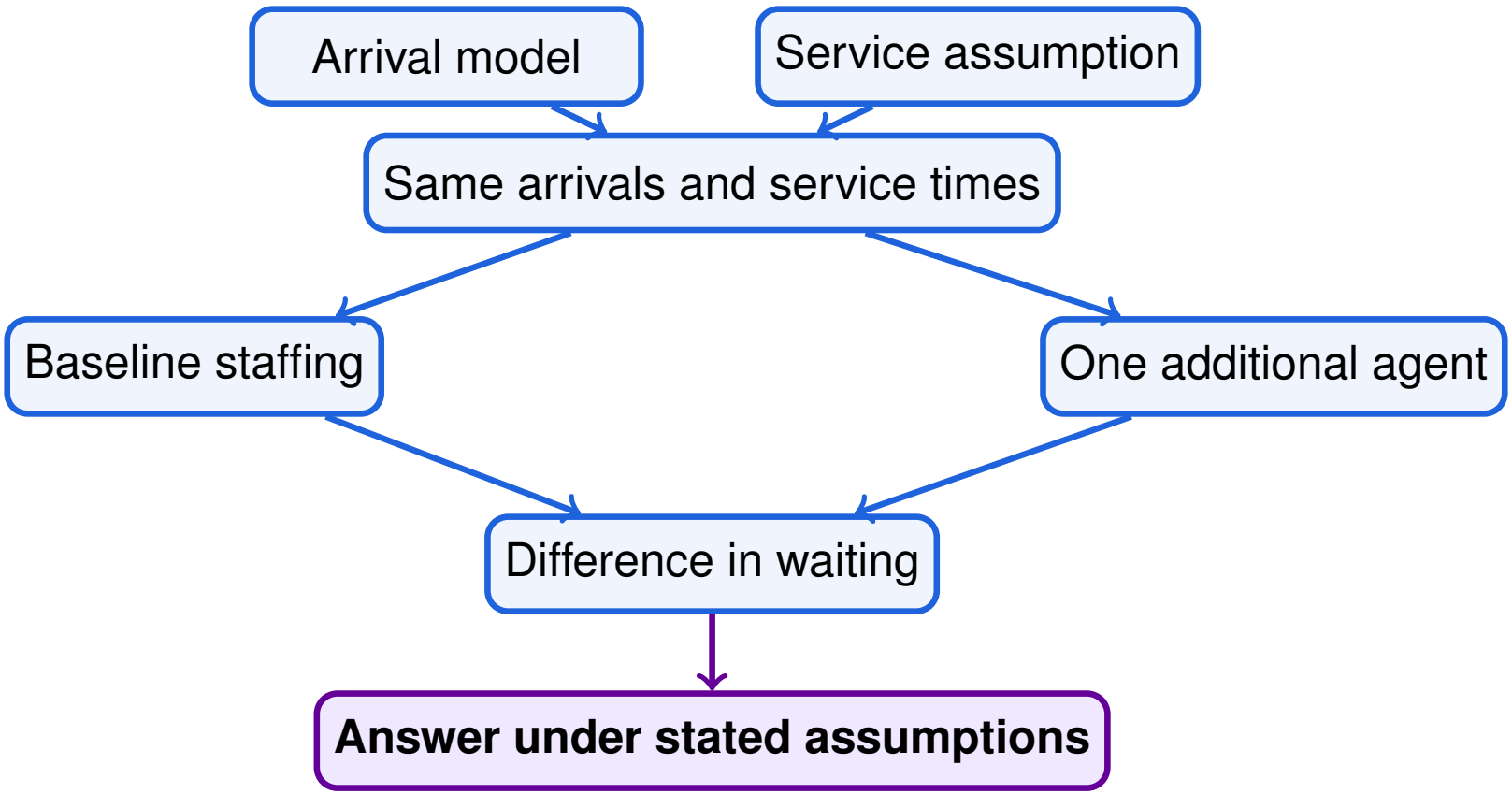
- One Poisson mean for each of the seven time periods.
- The observed counts vary more than this model predicts.

Candidate 2: variable days

- Multiply the seven means by one random daily demand level.
- This improves the variance fit during the morning and daytime.
- The evening variation remains unexplained.

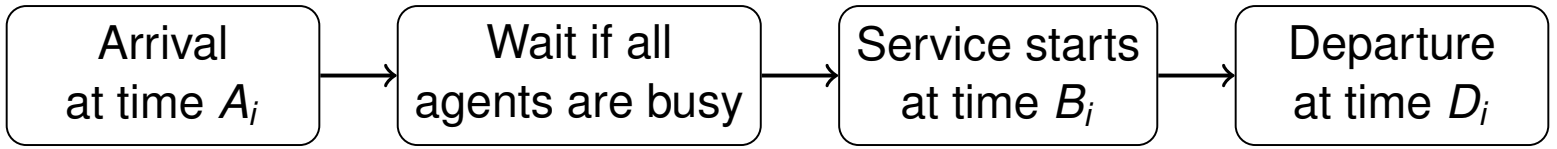
Conclusion: Candidate 2 is a provisional model not yet an “accepted” model for the complete day.

How much does one additional agent reduce waiting?



The queue changes when an event occurs

An **event** is an occurrence that changes the system: a call arrives, service starts, or service ends.



$$W_i = B_i - A_i, \qquad S_i = D_i - B_i, \qquad T_i = D_i - A_i.$$

- W_i: waiting time before service
- S_i: service duration
- T_i: total time in the modelled system

Start with a hand trace: one agent

Customer	Arrival A_i	Wait W_i	Start B_i	Service S_i	Departure D_i
1	0.0	0.0	0.0	2.0	2.0
2	1.0	1.0	2.0	2.0	4.0
3	2.5	1.5	4.0	1.0	5.0
4	2.7	2.3	5.0	2.5	7.5
5	3.0	4.5	7.5	1.5	9.0

$$B_i = \max(A_i, D_{i-1}), \qquad D_i = B_i + S_i.$$

Repeat the hand trace with two agents

Use the same arrivals A_i and service times S_i as before. Assign each customer to the agent who becomes free first.

Customer	A_i	W_i	B_i	S_i	D_i	Agent
1	0.0	0.0	0.0	2.0	2.0	1
2	1.0	0.0	1.0	2.0	3.0	2
3	2.5	0.0	2.5	1.0	3.5	1
4	2.7	0.3	3.0	2.5	5.5	2
5	3.0	0.5	3.5	1.5	5.0	1

Total waiting: 9.3 with one agent $\longrightarrow$ 0.8 with two agents.

What the simulator must record

Recorded value	What it lets us calculate
arrival time	demand chronology
service-start time	waiting time
departure time	time in system
assigned agent	workload of each agent
queue length after each event	queue length over time
number of busy agents after each event	agent occupancy over time

Summarize distributions, not only averages:

- **Waiting time across customers:** mean, median, quantiles and fraction who wait.
- **Queue length over time:** mean, quantiles, maximum and fraction of time the queue is non-empty.
- **Agent occupancy over time:** mean fraction of agents busy and fraction of time exactly $0, 1, \ldots, c$ agents are busy.

Check the implementation before the model

- Run both five-customer examples, then verify by hand that the code assigns the same agents, start times, waits and departure times.
- Check every customer: $B_i \geq A_i$, $D_i = B_i + S_i$, and $W_i \geq 0$.
- Increase service duration: waits should not decrease for the same arrivals.
- Run the same seed twice: the result should be identical.

Passing these checks supports the implementation. It does not show that the implementation is correct.

Make the checks repeatable with unit tests

A **unit test** is a small automatic check of one behaviour whose correct answer is already known.

```
import math

def test_two_agents():
    result = simulate_queue(
        arrivals=[0.0, 1.0, 2.5, 2.7, 3.0],
        services=[2.0, 2.0, 1.0, 2.5, 1.5],
        number_of_agents=2,
    )
    assert result["agent"] == [1, 2, 1, 2, 1]
    expected_waits = [0.0, 0.0, 0.0, 0.3, 0.5]
    pairs = zip(result["wait"], expected_waits)
    for got, expected in pairs:
        assert math.isclose(got, expected, abs_tol=1e-9)
```

Run the tests after every code change; add a test whenever a bug is found.

Inputs: evidence versus scenario

From the Week 1 extract

- time-of-day pattern of VRU entries
- strong differences between regular working days, Friday and Saturday
- pooled working-day periods and day-to-day demand variation

Scenario assumptions this week

- each selected VRU entry requests service
- exponential service duration
- identical agents and FIFO service
- no abandonment

Two generators chosen for the exercise

For one repetition, generate one arrival path from 09:00 to 11:00:

Parametric model

1. Draw one daily multiplier G_d.
2. Use
$$\lambda_{d,b} = \frac{G_d \widehat{m}_b}{15}$$
arrivals per minute.
3. Draw the time to the next arrival from $\text{Exponential}(\lambda_{d,b})$.

Whole-morning bootstrap

1. Sample one of the ten ordered vectors of eight morning counts, with replacement.
2. Keep the intervals in clock order.
3. Place the required arrivals uniformly inside each interval.

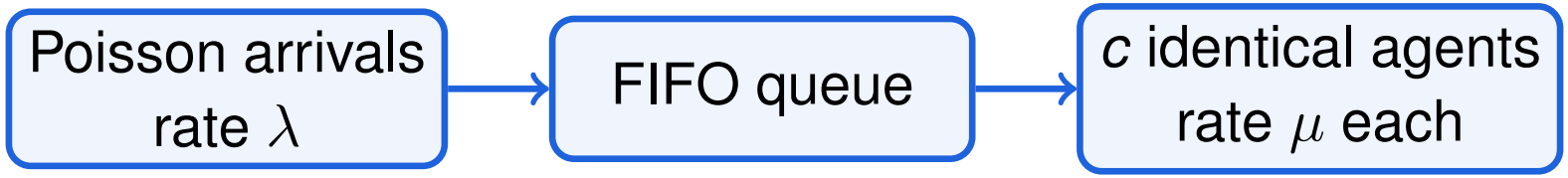
Service is a scenario input for now

Suppose service times are exponential with mean 6 minutes:

$$S_i \sim \text{Exponential}(\mu), \qquad \mathbb{E}[S_i] = 1/\mu = 6 \text{ minutes}.$$

- The value is chosen for a classroom scenario; it is not estimated from the Week 1 CSV.
- The exponential assumption makes a transparent first model.
- In later lectures we will estimate and compare service-time distributions using appropriate records.

A theoretical reference model: $M/M/c$



- The first *M*: independent exponential times between arrivals, equivalently a constant-rate Poisson arrival process.
- The second *M*: independent exponential service times with mean $1/\mu$.
- The c: the number of agents working in parallel.
- The waiting room is unlimited; service is FIFO; callers do not abandon.

This infinite-horizon model is a theoretical reference for the finite morning simulation.

The reference model gives a stability test

When a queue is present, all c agents are busy. Their combined service capacity is $c\mu$ calls per minute, while calls arrive at rate λ. Define

$$\rho = \frac{\lambda}{c\mu}.$$

For the infinite-horizon $M/M/c$ model, a stationary queue exists exactly when

$$\rho < 1 \qquad \text{or, equivalently,} \qquad \lambda < c\mu.$$

- If $\rho \geq 1$, arrivals match or exceed the maximum average service rate, so the queue cannot settle into a stationary distribution.
- If $\rho < 1$, the system has enough average capacity, but random arrival and service variation can still create long waits.

Apply the stability test to the morning scenario

For an average-demand morning, $\lambda = 39.9/15 = 2.66$ calls/min and $\mu = 1/6$ calls/min per agent.

Agents c	Capacity $c\mu$	ρ	Theoretical diagnosis
15	2.500	1.064	no stationary queue
16	2.667	0.998	stable, almost no spare capacity
17	2.833	0.939	stable, some spare capacity

Since $\lambda/\mu = 15.96$, 16 is the smallest stable integer for an average-demand morning; 17 is the first level with some buffer. These are candidate staffing levels, not yet a waiting-time conclusion. On a simulated morning, λ is multiplied by G_d.

First simulate around the capacity boundary

The theoretical model tells us where to look. Simulate the same arrival and service paths with:

15 agents 16 agents 17 agents.

- Does a queue remain at 11:00?
- How does the queue-length distribution change?
- How does the waiting-time distribution change?
- How often are all agents busy?

This locates the transition from insufficient capacity to useful spare capacity for the stated scenario.

Then measure the effect of one additional agent

After locating the critical range, compare 16 and 17 agents using identical arrivals and service requirements.

Report the change in:

- mean, median and upper quantiles of waiting time;
- fraction of callers who wait;
- queue-length distribution;
- agent-occupancy distribution;
- unfinished calls at 11:00.

What can we decide after Week 2?

Supported

- explain how limited capacity creates waiting;
- implement and check a multi-agent queue;
- compare staffing scenarios under stated inputs.

Still required for the bank

- agent-directed arrivals and routing;
- cleaned service-time data;
- staffing schedules and abandonment;
- uncertainty across repeated simulations.

Use an LLM to design stronger checks

Code-test prompt
Design three minimal tests for this FIFO multi-agent simulator. Include simultaneous arrivals, a tie between free agents and an invalid input. For valid cases, give every expected agent, start, wait and departure; for the invalid case, state the expected error.

Workflow-audit prompt
Trace the claim “17 agents reduce waiting relative to 16” through the arrival data, service assumptions, random generator, queue code and reported summaries. At each stage, propose one failure mode and one independent check.

Your work:

1. verify every predicted schedule by hand;
2. implement the valid examples as unit tests;
3. check every workflow claim against the data or stated assumptions;
4. record corrections to the LLM and retain the useful checks.

Notebook route

1. Reproduce the one-agent and two-agent hand traces.
2. Implement the rules for updating a multi-agent queue in Python.
3. Test invariants and simple limiting cases.
4. Reproduce the arrival-model fit and held-out checks.
5. Generate morning-peak arrivals with the parametric model and the whole-morning bootstrap.
6. Scan 15, 16 and 17 agents around the theoretical capacity boundary.
7. Compare 16 and 17 agents using the same random inputs.
8. State the conditional result and the evidence still missing.

What we take forward

- Waiting is an interaction between arrival timing, service duration and limited capacity.
- The $M/M/c$ stability condition identifies staffing levels that merit simulation.
- A short event-driven algorithm can simulate a FIFO multi-agent queue.
- Code checks and model checks answer different questions.
- A staffing comparison is only as credible as its arrival, service, routing and abandonment assumptions.

Next week: decide whether a simulated difference is larger than simulation noise.